

INDIVIDUAL REFLECTION

Working on this project gave me a better idea of how the concepts we learn in C++ can actually be used in a real application. At the beginning, the hotel reservation system looked like a simple program, but once we started designing it, there were many things to consider, such as guest details, room availability, reservations, services and billing.

I mainly understood the importance of dividing a large program into different classes. Instead of keeping everything in one place, we separated the different responsibilities of the system. This made the program easier to understand and also made it easier to find mistakes when something was not working properly.

The inheritance part was one of the areas I found useful. We used a common Room class and then created different room types from it. This helped me understand why inheritance is useful when several objects have some common properties but also have their own behaviour. The use of virtual functions and runtime polymorphism was another concept that became clearer while working on the project. Seeing different room objects being handled through a common base-class reference made the theory easier to understand.

I also spent time checking the program with different inputs. For example, we tested what happens when a user tries to book a room that is already occupied or enters an invalid option. These tests helped us notice problems in the program and correct them. I realized that writing the code is only one part of completing a project; testing it properly is equally important.

Working as part of a team was another useful experience. We divided the work into different areas and then combined the modules into one program. Sometimes a change in one part affected another part, so we had to communicate and test the complete program instead of checking only our individual sections.

Overall, this assignment helped me understand C++ OOP concepts in a more practical way. I became more comfortable with classes, inheritance, abstraction, polymorphism, constructors, destructors and pointers. More importantly, I learned how these concepts can work together to solve a practical problem. The project also improved my debugging, testing and teamwork skills, which I can use in future programming projects.